

## SAFETY DATA SHEET

Creation Date 24-January-2007

Revision Date 27-March-2024

Revision Number 4

### 1. Identification

**Product Name** 2,2,2-Trichloroethyl chloroformate

**Cat No. :** 97752

**CAS-No** 17341-93-4  
**Synonyms** beta,beta,beta-Trichloroethoxycarbonyl chloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                  |              |                            |
|------------------------------------------------------------------|--------------|----------------------------|
| <b>Acute oral toxicity</b>                                       | Category 4   |                            |
| <b>Acute Inhalation Toxicity</b>                                 | Category 3   | (based on evolved HCl gas) |
| <b>Skin Corrosion/Irritation</b>                                 | Category 1 B |                            |
| <b>Serious Eye Damage/Eye Irritation</b>                         | Category 1   |                            |
| <b>Specific target organ toxicity (single exposure)</b>          | Category 3   |                            |
| Target Organs - Respiratory system.                              |              |                            |
| <b>Health Hazards Not Otherwise Classified</b>                   | Category 1   |                            |
| In contact with water, releases gases which are toxic if inhaled |              |                            |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed

Toxic if inhaled

Causes severe skin burns and eye damage

May cause respiratory irritation

In contact with water, releases gases which are toxic if inhaled

**Precautionary Statements****Prevention**

Do not allow contact with water

Do not breathe dust/fumes/gas/mist/vapours/spray

Use only outdoors or in a well-ventilated area

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Rinse mouth

Do NOT induce vomiting

Wash contaminated clothing before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed

Store locked up

Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                         | CAS-No     | Weight % |
|---------------------------------------------------|------------|----------|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | >95      |

### 4. First-aid measures

|                     |                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>  | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.              |
| <b>Skin Contact</b> | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required. |
| <b>Inhalation</b>   | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Immediate medical attention is required.   |
| <b>Ingestion</b>    | Do NOT induce vomiting. Call a physician immediately.                                                                                          |

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                 |
|-----------------------------------------|-----------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                        |
| <b>Flash Point</b>                      | > 100 °C / > 212 °F                                             |
| <b>Method -</b>                         | No information available                                        |
| <b>Autoignition Temperature</b>         | No information available                                        |
| <b>Explosion Limits</b>                 |                                                                 |
| <b>Upper</b>                            | No data available                                               |
| <b>Lower</b>                            | No data available                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                        |
| <b>Sensitivity to Static Discharge</b>  | No information available                                        |

### Specific Hazards Arising from the Chemical

Contact with water liberates toxic gas. Water reactive. Produce flammable gases on contact with water.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                                                                                                                           |
| <b>Environmental Precautions</b>            | See Section 12 for additional Ecological Information.                                                                                                                                                                                                                 |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Wear self-contained breathing apparatus and protective suit. Do not let this chemical enter the environment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation. |
| <b>Storage.</b> | Corrosives area. Keep under nitrogen. Keep refrigerated. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong bases.                                           |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Physical State</b>                   | Liquid                                     |
| <b>Appearance</b>                       | Light yellow                               |
| <b>Odor</b>                             | pungent                                    |
| <b>Odor Threshold</b>                   | No information available                   |
| <b>pH</b>                               | No information available                   |
| <b>Melting Point/Range</b>              | 1 °C / 33.8 °F                             |
| <b>Boiling Point/Range</b>              | 171 - 172 °C / 339.8 - 341.6 °F @ 760 mmHg |
| <b>Flash Point</b>                      | > 100 °C / > 212 °F                        |
| <b>Evaporation Rate</b>                 | No information available                   |
| <b>Flammability (solid,gas)</b>         | Not applicable                             |
| <b>Flammability or explosive limits</b> |                                            |
| <b>Upper</b>                            | No data available                          |
| <b>Lower</b>                            | No data available                          |
| <b>Vapor Pressure</b>                   | 60 mmHg @ 75 °C                            |
| <b>Vapor Density</b>                    | 7.3 (Air = 1.0)                            |
| <b>Specific Gravity</b>                 | 1.539                                      |
| <b>Solubility</b>                       | Decomposes in contact with water           |

|                                        |                          |
|----------------------------------------|--------------------------|
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C3 H2 Cl4 O2             |
| Molecular Weight                       | 211.86                   |

## 10. Stability and reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                      |
| Stability                        | Moisture sensitive.                                                                      |
| Conditions to Avoid              | Incompatible products. Exposure to moist air or water.                                   |
| Incompatible Materials           | Strong oxidizing agents, Strong bases                                                    |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Phosgene, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                         | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mutagenic Effects                          | No information available                                                                                                                                                                                                                                   |
| Reproductive Effects                       | No information available.                                                                                                                                                                                                                                  |
| Developmental Effects                      | No information available.                                                                                                                                                                                                                                  |
| Teratogenicity                             | No information available.                                                                                                                                                                                                                                  |
| STOT - single exposure                     | Respiratory system                                                                                                                                                                                                                                         |
| STOT - repeated exposure                   | None known                                                                                                                                                                                                                                                 |
| Aspiration hazard                          | No information available                                                                                                                                                                                                                                   |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| Endocrine Disruptor Information            | No information available                                                                                                                                                                                                                                   |

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

**UN-No** UN3277  
**Proper Shipping Name** Chloroformates, toxic, corrosive, n.o.s.  
**Technical Name** Carbonochloridic acid, 2,2,2-trichloroethyl ester  
**Hazard Class** 6.1  
**Subsidiary Hazard Class** 8  
**Packing Group** II

### TDG

**UN-No** UN3277  
**Proper Shipping Name** Chloroformates, toxic, corrosive, n.o.s.  
**Hazard Class** 6.1  
**Subsidiary Hazard Class** 8  
**Packing Group** II

### IATA

**UN-No** UN3277  
**Proper Shipping Name** Chloroformates, toxic, corrosive, n.o.s.  
**Hazard Class** 6.1  
**Subsidiary Hazard Class** 8  
**Packing Group** II

### IMDG/IMO

**UN-No** UN3277  
**Proper Shipping Name** Chloroformates, toxic, corrosive, n.o.s.  
**Hazard Class** 6.1  
**Subsidiary Hazard Class** 8  
**Packing Group** II

## 15. Regulatory information

### International Inventories

| Component                                         | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | -   | X    | X    | ACTIVE                                        | 241-363-7 | -      | -   |

| Component                                         | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | -     | KE-05693 | X    | X    | X    | X    | X     | -     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                         | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                         | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Carbonochloridic acid, 2,2,2-trichloroethyl ester | 17341-93-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Product Safety Department  
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[www.thermofisher.com](http://www.thermofisher.com)

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**